

Answer Key — Biology Grade 11

Sub-Strand 1.4: Cell Division • Lesson 3: Types of Cell Division: Mitosis and Its Stages

Teacher reference only. Do not project or hand out.

Quick marking grid

Q1	Q2	Q3	Q4	Q5	Q6	Q7
C	D	B	C	A	C	C

USE AFTER: PREDICT PHASE — Predict: individual sketch/writing on what happens to the chromosomes when one cell becomes two

Q1. A skin cell in a growing child has 4 chromosomes. It divides by mitosis into two new cells. How many chromosomes will each new cell have?

- A. A different number in each cell, because new cells get new instructions
- B. 2 chromosomes, because the chromosomes are split in half between the cells
- C. 4 chromosomes, the same as the original cell**
- D. 8 chromosomes, because each new cell makes extra instructions

Correct answer: C. In mitosis the chromosomes are copied and then shared equally, so each daughter cell has the same number (4) as the parent cell. Halving the chromosomes or making new instructions are common misconceptions.

USE AFTER: OBSERVE PHASE — Part 2 (Research): matching the terms to the observations, 'We observed _____. This is called _____.'

Q2. Learners studying a dividing onion root tip cell see the chromosomes lined up along the middle (equator) of the cell. Which stage of mitosis is this?

- A. Prophase
- B. Telophase
- C. Anaphase
- D. Metaphase**

Correct answer: D. Chromosomes lining up at the equator of the cell is metaphase. In prophase they are only becoming visible, in anaphase they separate, and in telophase two nuclei form.

USE AFTER: OBSERVE PHASE — Part 2 (Research): matching the terms to the observations, 'We observed _____. This is called _____.'

Q3. In a dividing cell, the chromosomes have reached opposite ends of the cell and two new nuclei are forming. Which stage of mitosis is this?

- A. Metaphase, because the chromosomes are in an ordered position
- B. Telophase, because two new nuclei form around the chromosomes**
- C. Anaphase, because the chromosomes are moving apart
- D. Prophase, because the chromosomes are visible

Correct answer: B. Two nuclei forming at the opposite ends of the cell is telophase. Anaphase is the earlier stage in which chromosomes are still moving apart.

USE AFTER: EXPLAIN PHASE — Return to the phenomenon: 'How does mitosis help a child grow?' and the Discussion Diamond

Q4. A newborn baby weighing 3 kg grows into a teenager of 50 kg. Which statement best explains how mitosis helps this child grow?

- A. Cells only get bigger, so mitosis is not needed for growth
- B. Mitosis splits the chromosomes in half so each new cell gets only part of the instructions
- C. Mitosis copies and shares chromosomes accurately to make two daughter cells, so the number of cells increases and the body grows**
- D. Mitosis makes new cells that are different from the parent cell so the body can have many cell types

Correct answer: C. One cell divides into two daughter cells with the same chromosomes as the parent cell. More cells means growth and repair. Growth is not only cells getting bigger, and the chromosomes are not halved.

USE AFTER: EXPLAIN PHASE — Discussion Diamond: statement 'Eating food makes new cells by itself' and the teacher question 'what process makes the new cell?'

Q5. A child in Kericho eats ugali, sukuma wiki and beans and grows taller. Which statement correctly links the food to the growth of new cells?

- A. Food gives energy and building materials, but new cells are made when existing cells divide by mitosis**
- B. Food is not needed because mitosis makes new cells from nothing
- C. Food makes the chromosomes inside cells change into new instructions
- D. Food turns directly into new cells without any cell division

Correct answer: A. Food supplies energy and materials, but the process that makes new cells is cell division by mitosis. Food alone cannot produce cells.

USE AFTER: DQB PHASE — Exit ticket: write the four stages in order, plus cytokinesis

Q6. Which list shows the stages of mitosis in the correct order, followed by the division of the cytoplasm?

- A. Prophase, metaphase, telophase, anaphase, cytokinesis
- B. Metaphase, prophase, anaphase, telophase, cytokinesis
- C. Prophase, metaphase, anaphase, telophase, cytokinesis**
- D. Prophase, anaphase, metaphase, telophase, cytokinesis

Correct answer: C. The order is prophase, metaphase, anaphase and telophase, then cytokinesis divides the cytoplasm to make two daughter cells.

USE AFTER: MODEL BUILDING PHASE — Cumulative model: labelled flow diagram from parent cell to two identical daughter cells

Q7. A group draws a flow diagram of mitosis. After the two nuclei have formed, the last step shows the cell dividing into two separate daughter cells. What is this step called?

- A. Prophase
- B. Anaphase
- C. Cytokinesis**
- D. Telophase

Correct answer: C. Cytokinesis is the division of the cytoplasm that makes two separate daughter cells after the four stages of mitosis. Telophase is when the two nuclei form.